

Name: _____

Total: 16 points

We will consider a 2D Ornstein-Uhlenbeck process given by the following SDE:

$$dX_t = -\theta X_t dt + \sigma dW_t, \quad (1)$$

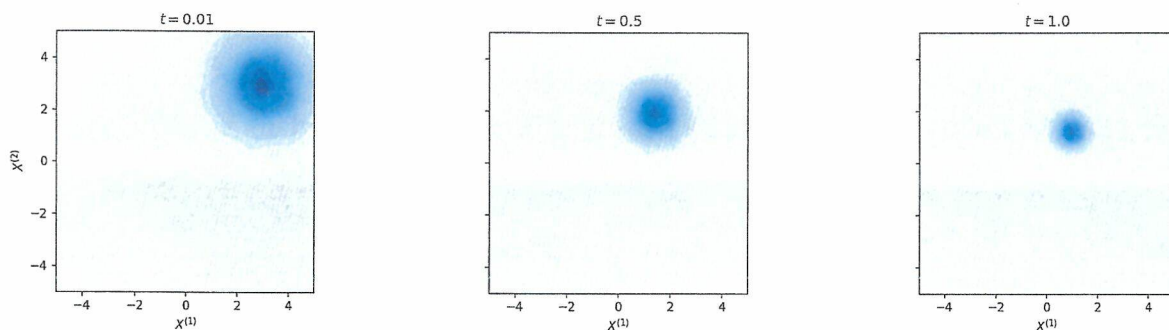
where $\theta, \sigma > 0$ are constants. We will treat the paths $\{X_t\}$ as the solution of a random dynamical system, where the randomness comes from the standard Wiener process, W_t , in 2D. There is a unique, invariant measure of the process given by

$$X_\infty \sim \mathcal{N}\left(0, \frac{\sigma^2}{2\theta} I_2\right). \quad (2)$$

We can use Ito integrals to obtain that the process X_t is the following random variable,

$$X_t = e^{-\theta t} X_0 + \sigma \int_0^t e^{-\theta(t-s)} dW_s. \quad (3)$$

- Below we see a simulation of the densities (estimated) of sample paths for a fixed Wiener path. That is, for a fixed path W , we see the distribution $\varphi_{W\#}^t \rho_0$ for some initial density, ρ_0 (a shifted Gaussian), at different times t . As t increases, the densities will concentrate at a single point. True or False? Justify your answer. (2 points)



2. What is the distribution of the limiting points/set of points obtained from such simulations with fixed Wiener paths? In other words, what is $\lim_{t \rightarrow \infty} \mathbb{E}_{W \sim \mathbb{P}} \varphi_{W_t}^t \rho_0$, where $\mathbb{P}$ is the Wiener measure? (2 points)

3. Let $P_t(x, x_0)$ be the transition probability density of $X_t = x$ conditioned on $X_0 = x_0$. The Koopman operator $U^t : L^2(\mu) \rightarrow L^2(\mu)$ is given by

$$U^t f(x_0) = \int P_t(x, x_0) f(x) dx. \quad (4)$$

Relate this definition to the Koopman operator for the deterministic system, φ_W^t . (2 points)

4. What is the Frobenius-Perron operator associated with the Koopman semigroup U^t ? Write down how it acts on probability densities. *Hint: it is the $L^2(dx)$ adjoint of U^t .* (2 points)
5. Let $H_n(x) = h_{n_1}(x^{(1)}) h_{n_2}(x^{(2)})$ be a product of 1D Hermite polynomials. Here, $n = (n_1, n_2)$ is a tuple of non-negative integers and $(x^{(1)}, x^{(2)})$ are coordinates of a point x . These are eigenfunctions of U^t with eigenvalues $e^{-\theta(n_1+n_2)t}$. Knowing this, what do you expect the Koopman approximation to look like from EDMD, when choosing the span of H_n as the basis? (3 points)
6. Briefly describe how you would perform EDMD for this system using the basis functions from the previous part. (3 points)

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7. Any initial density converges to a Gaussian exponentially fast. Argue why or why not. (2 points)